Project Title: IndicSent-M: Multimodal Sentiment Analysis for Indian Languages Using Text and Images

Motivation:

The rapid growth of social media platforms has resulted in a large amount of user-generated content in Indian languages. Social media posts, memes, and online reviews frequently combine **text and visual information** to express opinions and emotions. Traditional sentiment analysis systems mainly focus on text and often fail to capture sentiment expressed through images, memes, emojis, cultural references, and the interaction between text and visual content.

Furthermore, sentiment analysis for Indian languages is challenging due to **linguistic diversity, limited resources, code-mixed content, variations in scripts, and cultural expressions**. Existing multilingual models may perform well for high-resource languages but can struggle with low-resource Indian languages.

This project aims to develop a **multimodal sentiment analysis system** that jointly processes textual and visual information from Indian-language social media content. The system will investigate whether combining text and image representations can improve sentiment classification compared with text-only and image-only approaches. The project will also explore explainable predictions to help users understand which textual and visual features contributed to the final sentiment prediction.

Problem Statement:

Current sentiment analysis systems primarily focus on **text-only data** and are often developed for high-resource languages such as English. Such systems are inadequate for Indian social media content where users frequently communicate through **memes, images, multilingual text, code-mixed language, emojis, and culturally specific expressions**.

In multimodal social media content, the sentiment expressed by the text may differ from or complement the sentiment conveyed by the image. For example, a positive-looking sentence may be used with a negative image to express sarcasm or criticism. Therefore, analyzing only one modality may lead to incorrect sentiment predictions.

Proposed Scope:

The project will develop a **multimodal sentiment classification system for Indian-language memes/social media content** using both text and images.

The proposed system will include:

1. **Text Processing:** Cleaning, normalization, language identification, and preprocessing of Indian-language text.
2. **Text Representation:** Extraction of semantic features using multilingual Transformer models such as **IndicBERT or XLM-R**.
3. **Image Representation:** Extraction of visual features using pretrained models such as **ResNet or Vision Transformer (ViT)**.
4. **Multimodal Fusion:** Combining textual and visual representations using feature concatenation or attention-based fusion.
5. **Sentiment Classification:** Classification into **Positive, Neutral, and Negative** categories.

6. **Comparative Analysis:** Comparison of:
    - Text-only model
    - Image-only model
    - Text + Image multimodal model
7. **Explainability:** Investigating important words/regions contributing to sentiment predictions using suitable XAI techniques.
8. **Evaluation:** Performance evaluation using **Accuracy, Precision, Recall, F1-score, and Confusion Matrix**.

Expected Outcome:

The project is expected to demonstrate whether **multimodal learning improves sentiment classification** compared with individual modalities. The final system should be able to accept an Indian-language meme containing text and an image and predict its sentiment.

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Deliverables Checklist:

- **Source Code**
- **Dataset Preparation & Documentation**
- **Trained Multimodal Model**
- **Research Paper**
- **Presentation Slides**